FOCS_StudF03: Student's Progress Report

Tunku Abdul Rahman University of Management and Technology
Faculty of Computing and Information Technology
Industrial Training Progress Report

Activity Log



Name of Trainee:	Yeap Zi Jia
Name of Company:	SOPHIC AUTOMATION SDN BHD.
Month/Year	MAY/2026
	(E.g. JAN/2022)

Week	Projects / Activities
1	**Description of project/activity** **Activity 1: Excel API Development in VDI Environment**Reviewed the Developed backend API in VDI environment for Excel import and export functionality. Implemented endpoints for file upload, data processing, and export, and performed integration testing with external APIs. **A total of Time spent on the activity** • Activity 1: Excel API Development is 9 hours **Technology/application used to complete the activity** • ASP.NET Web API (C#) • VDI Environment • Visual Studio • Excel processing libraries • API testing tools (Postman / Swagger) **What challenges have you faced (if any)?** Understanding differences between .NET Framework and .NET Core required additional research. Working in the VDI environment also posed limitations in configuration and access, which affected development setup.

	What skills/knowledge have you learned from the activities? Gained better understanding of framework differences and improved backend API development skills, especially in Excel handling and API integration. Also enhanced troubleshooting skills in restricted environments.
2	**Description of project/activity** **Activity 1: FDC Background Scheduler Stabilization** Configured and stabilized the FDC background execution loop to ensure continuous processing without manual intervention. Resolved issues related to IIS application pool suspension and timeout behavior by improving scheduler persistence, monitoring, and recovery handling. **Activity 2: Data Extraction Logic Refactoring** Refactored and modularized the API data pulling logic to improve code maintainability and readability. Enhanced the handling of tool and chamber hierarchy mapping to ensure accurate summary data generation and more reliable external API integration. **Activity 3: Database Insert & Update Optimization** Optimized the database synchronization workflow by restructuring insert and update processes. Updated Entity Framework mappings to align with revised Oracle database schemas and resolved broken database triggers to ensure successful data synchronization operations. **A total of Time spent on the activity** • Activity 1: FDC Background Scheduler Stabilization is 15 hours • Activity 2: Data Extraction Logic Refactoring is 17 hours • Activity 3: Database Insert & Update Optimization is 13 hours **Technology/application used to complete the activity** • ASP.NET Web API (C#) • Entity Framework • Oracle Database

- IIS Web Server
- Visual Studio
- VDI Environment

What challenges have you faced (if any)?

Faced challenges in maintaining persistent background execution within the IIS hosting environment, as the application pool suspension and timeout limits occasionally caused the scheduler to stop unexpectedly, requiring manual restarts.

Additionally, the original API data pulling logic was tightly coupled and difficult to maintain, especially when processing complex tool and chamber hierarchy structures from external API responses. This occasionally resulted in inaccurate summary data generation and inconsistent request handling.

Database synchronization also presented challenges due to outdated Entity Framework mappings and broken Oracle database triggers, which caused insertion and update failures during synchronization operations.

What skills/knowledge have you learned from the activities?

Gained deeper understanding of IIS application pool lifecycle behavior, timeout configurations, and strategies for maintaining stable background execution processes in web applications.

Improved skills in backend architecture refactoring by transforming complex monolithic data extraction logic into smaller, modular, and maintainable components that simplify debugging and future enhancements.

Enhanced practical experience in Oracle database troubleshooting, particularly in resolving broken triggers, schema alignment issues, and Entity Framework mapping inconsistencies.

	Also developed stronger knowledge in designing resilient synchronization workflows, including monitoring, logging, and error-handling mechanisms to improve system reliability and maintainability.
3	**Description of project/activity** **Activity 1: QA Functional Testing** Performed quality assurance (QA) testing on implemented system functions to verify that workflows, business logic, and system processes were operating correctly according to functional requirements. Conducted validation of user flows and identified inconsistencies during testing to ensure system stability and reliability. **Activity 2: Excel Import Validation Enhancement** Enhanced the Excel import module by implementing additional validation rules to improve data integrity and prevent invalid or incomplete records from being uploaded into the system. Added validation checks for required fields, invalid formats, and incorrect data structures to strengthen the import process. **Activity 3: API Testing Function for Nintex K2 Integration** Developed and enhanced backend API testing functions in the C# backend system to support integration validation for Nintex K2 workflow triggering. Assisted in verifying API accessibility, request handling, and response validation to ensure reliable communication between systems. **A total of Time spent on the activity** • Activity 1: QA Functional Testing is 8 hours • Activity 2: Excel Import Validation Enhancement is 10 hours • Activity 3: API Testing Function for Nintex K2 Integration is 12 hours **Technology/application used to complete the activity** • ASP.NET Web API (C#) • Visual Studio • SQL Database • Excel File Processing

- Nintex K2 Integration
- VDI Environment

What challenges have you faced (if any)?

Faced challenges in handling inconsistent and incomplete Excel data during the import process, as invalid formats and missing required fields could potentially cause processing failures and inaccurate data storage within the system.

Additionally, validating API integration for Nintex K2 workflow triggering required careful verification of request payload structures, endpoint accessibility, and response handling to ensure seamless communication between external workflow systems and backend services.

Ensuring all implemented system functions operated correctly during QA testing also required detailed functional verification and troubleshooting to identify unexpected validation or workflow-related issues.

What skills/knowledge have you learned from the activities?

Gained practical experience in performing QA functional testing, including validating business workflows, identifying defects, and verifying system functionality against requirements.

Improved understanding of backend validation techniques by implementing robust Excel import validation logic to enhance data integrity and prevent invalid system transactions.

Enhanced knowledge of API integration concepts, particularly in backend API testing, request and response validation, and external workflow system integration using Nintex K2.

Also strengthened debugging and troubleshooting skills while analyzing validation failures, resolving API communication issues, and improving overall backend system reliability and maintainability.

4	**Description of project/activity** **Activity 1: Batch Modify UI Enhancement (DEV & UAT)** Improved and refined the user interface (UI) for the Batch Modify page in both DEV and UAT environments to enhance usability, layout consistency, and overall user experience. Adjusted UI components and page structure to improve readability and simplify user interaction during batch modification processes. **Activity 2: Excel Format Update in C# Backend** Updated and enhanced the Excel import/export processing logic within the C# backend application to support the latest required file structure. Modified the backend handling process to ensure compatibility with updated Excel templates and improve overall data processing reliability. **Activity 3: API Project Conversion & Windows Task Scheduler Research** Conducted research and technical exploration on converting the existing API project into a Console Application (.exe) for scheduler-based execution. Investigated Windows Task Scheduler configuration, execution flow, and deployment setup to support automated background task execution. **A total of Time spent on the activity** • Activity 1: Batch Modify UI Enhancement (DEV & UAT) is 15 hours • Activity 2: Excel Format Update in C# Backend is 12 hours • Activity 3: API Project Conversion & Windows Task Scheduler Research is 18 hours **Technology/application used to complete the activity** • ASP.NET Web API (C#) • Visual Studio • Windows Task Scheduler • SQL Database • Excel File Processing

- DEV & UAT Environment
- VDI Environment

What challenges have you faced (if any)?

Faced challenges in maintaining UI consistency across DEV and UAT environments, as layout differences and component alignment issues affected usability and data presentation on the Batch Modify page.

Additionally, updating the Excel processing format required careful alignment between the latest template structure and the existing backend processing logic to avoid compatibility issues and data processing failures.

Researching the conversion of the API project into a Console Application also required understanding how scheduled background execution interacts with Windows Task Scheduler configuration, deployment setup, and automated triggering mechanisms.

What skills/knowledge have you learned from the activities?

Gained practical experience in UI enhancement and environment-based deployment verification by improving the Batch Modify page across DEV and UAT environments.

Enhanced understanding of Excel processing workflows within C# backend applications, including updating import/export logic and maintaining compatibility with evolving template structures.

Improved knowledge of Console Application development and Windows Task Scheduler configuration for automated backend execution and scheduled task management.

Also strengthened research, troubleshooting, and problem-solving skills while exploring scheduler-based execution strategies, deployment configurations, and background automation approaches for enterprise applications.

5	**Description of project/activity** **Activity 1: RTM System Excel Import & Export Swagger API** Developed and configured the Swagger API for Excel import and export functionality in the RTM system to support file upload, data processing, and downloadable report generation processes. Enhanced API request and response validation to ensure compatibility with the required Excel file structure and improve reliability during data handling operations. **Activity 2: Meeting Minutes Documentation** Prepared and organized meeting minutes based on project discussions, task progress updates, and action items. Structured the meeting documentation clearly to improve communication, task tracking, and future reference for the development team and stakeholders. **A total of Time spent on the activity** • Activity 1: RTM System Excel Import & Export Swagger API is 30 hours • Activity 2: Meeting Minutes Documentation is 6 hours **Technology/application used to complete the activity** • ASP.NET Web API (C#) • Swagger API Documentation • Visual Studio • Excel File Processing • RTM System • Microsoft Office **What challenges have you faced (if any)?** Faced challenges in ensuring the Excel import and export API handled different file structures and validation requirements correctly to maintain compatibility with the RTM system workflow. Additional verification and testing were required to ensure successful file upload, processing, and export functionality through Swagger API testing.

	Another challenge encountered was organizing meeting discussion points and action items into clear and structured meeting minutes while ensuring all important project updates and follow-up tasks were accurately documented for team reference. **What skills/knowledge have you learned from the activities?** Gained practical experience in developing and testing Excel import and export APIs using Swagger within the RTM system environment. Improved understanding of API request and response validation, file processing workflows, and backend handling for Excel-based operations. Enhanced documentation and communication skills through preparing structured meeting minutes and summarizing technical discussions effectively. Also strengthened problem-solving and testing skills while validating API functionality and handling data processing requirements.

Suggestions / Comments / Additional information (if any):

Leave Application / Leave Taken

1.From (dd/mm/yyyy): 05/05/2026 to (dd/mm/yyyy) 05/05/2026 (0.5 day(s))
2. Reasons for taking leave: Personal matter
3. Total number of days taken: 0.5

I hereby declare that the information given above is correct.

Signature: Yeap **Date** (dd/mm/yyyy): 29/05/2026

Endorsement by the Company Supervisor:

The above is a true record of activities taken by the trainee in the captioned week.

Signature of Supervisor: 

Name of Supervisor:	Teoh Wei De
Date *(dd/mm/yyyy)*:	29/05/2026
Email:	weide.teoh@sophicautomation.com
Mobile / Office Contact No.:	019 - 466 1839

Company Stamp:

